

Rajvardhan Patil

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EDUCATION

University of Colorado Boulder

Master of Science in Computer Science

Coursework: Neural Networks and Deep Learning, Big Data Architecture, Database Systems, Enterprise Networks, Quantum Computing

Boulder, CO, USA

August 2025 – May 2027

Vellore Institute of Technology (VIT)

Bachelor of Technology in Computer Science and Engineering; GPA: 3.4/4.0

Vellore, India

July 2021 – May 2025

EXPERIENCE

University of Colorado Boulder – Institute of Cognitive Science

Software Engineer (Student Assistant)

Boulder, CO, USA

May 2026 – Present

- Operate the UVI **Flask/MongoDB** platform serving **4 linguistic resources** (VerbNet, PropBank, FrameNet, OntoNotes) to NLP researchers worldwide; triaged and resolved critical production defects to sustain **99.99% uptime**, verified by **18,772 automated health checks** rather than self-reported status – an **evidence-first** approach to service reliability.
- Designed a **shared behavioral representation** across **4 heterogeneous corpora**: built an automated **NLP ETL pipeline** (**spaCy** dependency parsing, **NLTK** tokenization) that mines, **normalizes**, and cross-links inconsistent source annotations into structured, semantically comparable MongoDB collections powering **REST** search endpoints.
- Cut page-load and search **latency** across **20+ views** through query, index, and Jinja2 render optimization; administer **2 production Linux servers** with **zero-downtime** gunicorn deployments, **OpenSSL CVE** remediation from Qualys scans, and a **Flask 3 / Python 3.12** stack upgrade executed without a single service interruption.

University of Colorado Boulder – Civil & Architectural Engineering

Software Engineer (Student Assistant)

Boulder, CO, USA

March 2026 – Present

- Architected **SABER (link)**, a **full-stack** energy audit platform: **FastAPI/Python** backend on **Azure** exposing **asynchronous REST APIs** over a change-point regression engine, with an **LLM/RAG** pipeline that returns **no unsupported answer** – every standards citation traces back to a retrieved source passage – plus a **React/TypeScript** frontend with **Plotly** visualizations.
- Built **Panel Sizer (link)**, a **containerized FastAPI** NEC Article 220 load calculator deployed on **Azure Container Apps** with **auto-scaling** and an integrated **RAG** assistant, replacing a manual spreadsheet workflow with a repeatable, **API-driven microservice**.
- Developed the **TBEEC Compliance Tool**, an **LLM/RAG** service on **Azure** that automatically **compares submitted designs against energy-code text** and explains which requirements are met, partially met, or unaddressed; packaged into **2 distribution formats** (installable Python library and Windows executable) for consultant use.

Computer Association of Kolhapur (CAK)

Software Developer (Volunteer)

Kolhapur, India (Remote)

January 2026 – Present

- Replaced a fully manual, paper-based membership process with a **full-stack** portal (**Next.js**, **FastAPI**, **PostgreSQL**) featuring **multi-step digital onboarding**, **Razorpay** payment integration, automated **PDF receipt** generation, and **event-driven** membership activation on payment confirmation.
- Delivered a **JWT-secured** admin dashboard with **real-time** membership and revenue reporting, plus a bulk **Excel import** pipeline that parses, validates, and deduplicates records before insertion, producing a **structured audit trail** of every membership state change.

Walstar Technologies

Software Engineering Intern

Kolhapur, India

August 2023 – October 2023

- Digitally transformed a manual, paper-driven dairy operation into an end-to-end app (**Flutter**, **Laravel**, **MySQL**) adopted by **5+ cooperatives**, cutting manual operations **40%** and saving each cooperative roughly **7 hours per week**.
- Built **REST APIs** in **PHP/MySQL** sustaining **99.8% uptime** under test load; prototyped UI/UX in **Figma** and shipped an Android frontend validated at **95% UI test coverage**.

PROJECTS

- TopoPulse**: **Python-based network monitoring and management platform** for a **Containerlab-emulated Cisco IOS topology**, polling device telemetry via **SSH/Netmiko** and streaming health metrics into a **Grafana dashboard** for real-time visualization. Automated configuration and validation of **Cisco proprietary protocols** – **VPC**, **HSRP**, **GLBP**, and **CDP** – for redundant gateway failover and topology discovery. Tech: **Python**, **Containerlab**, **Cisco IOS**, **Grafana**, **Prometheus**, **Netmiko**.
- ClearRail**: **Fault-tolerant payment gateway and processor** routing verified, encrypted requests across independent **Authorization**, **Clearing**, and **Settlement** microservices behind a load balancer, with **async fraud detection**. Load-tested for **sustained throughput**, **autoscaling**, **DOS resilience**, and **zero dropped transactions on single-service failure**. Tech: **Java**, **REST/gRPC/Kafka**, **JDBC**, **Docker**.
- QueueWright**: **Django workflow platform with an AI quality-review agent** that pulls items from a **QA queue**, executes the QA workflow, runs a **quality check**, and assigns to a human instead of auto-approving. Reviewer edits are approver-gated and logged as a **structured audit trail** (client, entries, users). Tech: **Python**, **Django**, **LLM agents**, **PostgreSQL**, **Celery**.

SKILLS SUMMARY

- Languages**: **Python**, **Java**, **JavaScript**, **TypeScript**, **SQL**, **PHP**, **HTML**, **CSS**.
- Frameworks**: **React**, **Next.js**, **Django**, **FastAPI**, **Flask**, **Node.js**, **Express**, **SQLAlchemy**, **Hibernate**, **Redux**, **Flutter**, **Laravel**, **Tailwind CSS**.
- Networking**: Routing & Switching (**OSPF**, **EIGRP**, **RIPv2**), **VLANs/Trunking**, **STP/RSTP**, **IPSec** & **VPN** (**DMVPN**), **MPLS**, **NAT/ACLs**, **IPv4/IPv6 subnetting**, **Wireshark** packet analysis, multi-router lab topologies, **Cisco IOS/Junos** troubleshooting.
- Distributed Systems**: **Microservices**, **REST/gRPC/GraphQL**, **Kafka**, load balancing, **fault tolerance** & failover, **API gateways**, **WebSockets**, load & stress testing.
- Databases & Cloud**: **PostgreSQL**, **MongoDB**, **MySQL**, **Redis**, **Supabase**, **Firebase**, **AWS** (**S3**, **CloudFront**, **Lambda**), **Azure** (**App Service**, **Container Apps**), **Docker**, **Linux** administration.
- AI/ML**: **LLMs** (**Claude**, **GPT**, **Ollama**), **LangChain**, **RAG**, **AI agents**, vector databases (**ChromaDB**), prompt engineering, **TensorFlow**, **scikit-learn**, **spaCy**, **NLTK**.
- Tools & Practices**: **Git**, **GitHub Actions**, **CI/CD**, automated testing & coverage analysis, **Grafana**, **PostHog**, **Postman**, **Figma**, **Agile/SCRUM**.